Walmart Retail Sales Forecasting & Business Analytics Dashboard

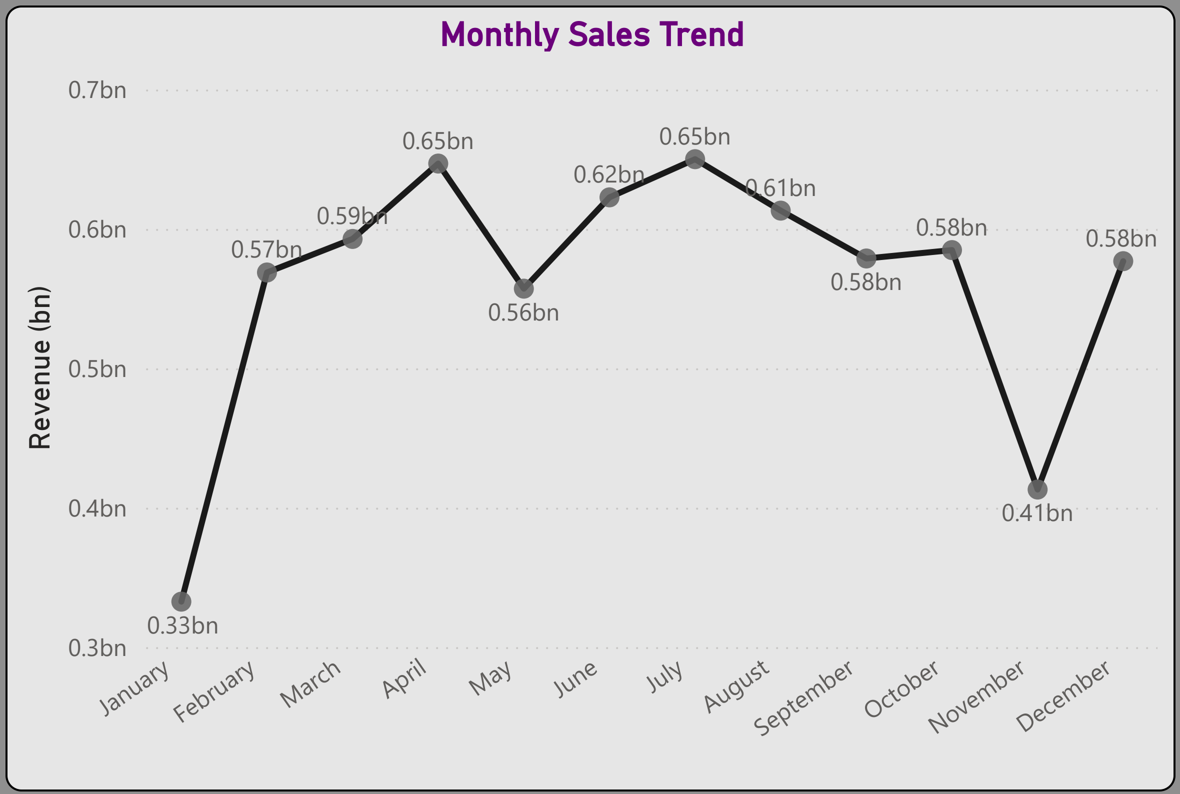
Interactive Sales Performance & Forecasting Dashboard

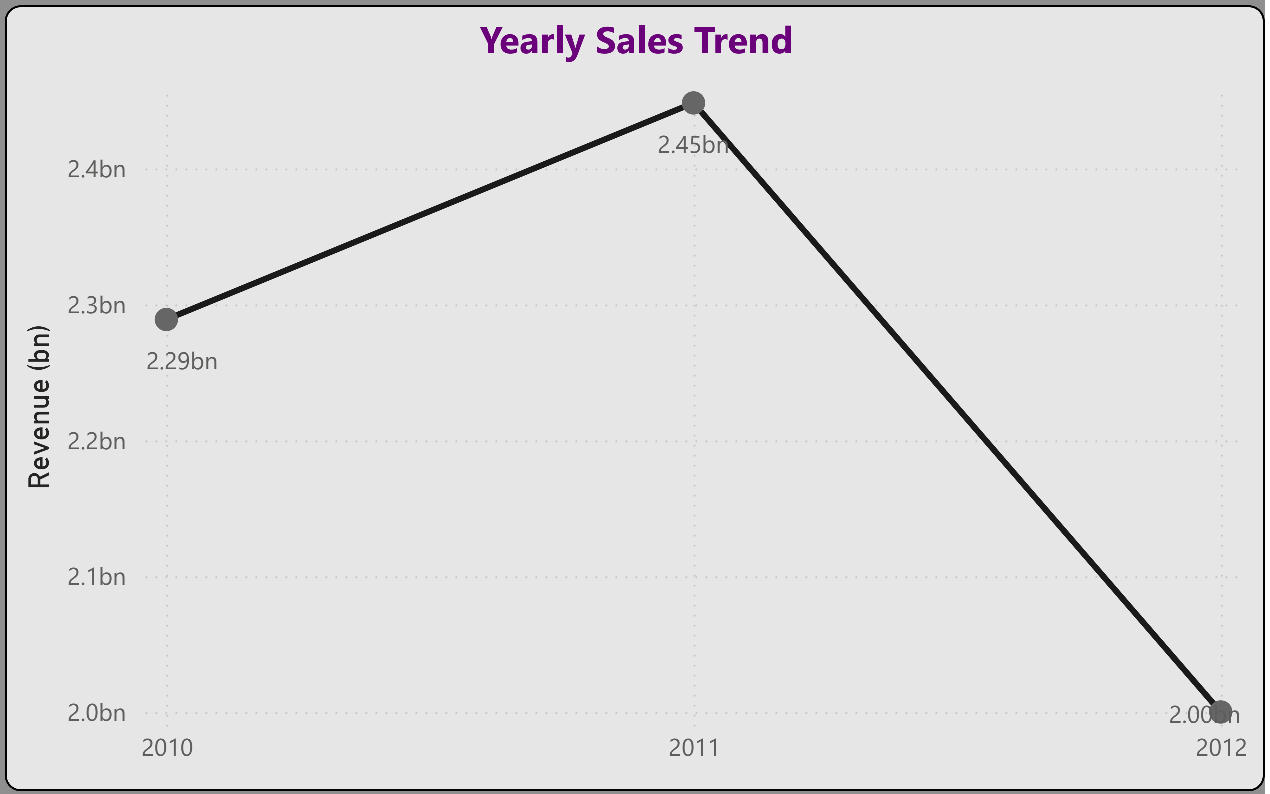
6.74bn
Total Revenue

16.03K
Average Weekly Sales

45
Total Stores

81
Total Departments

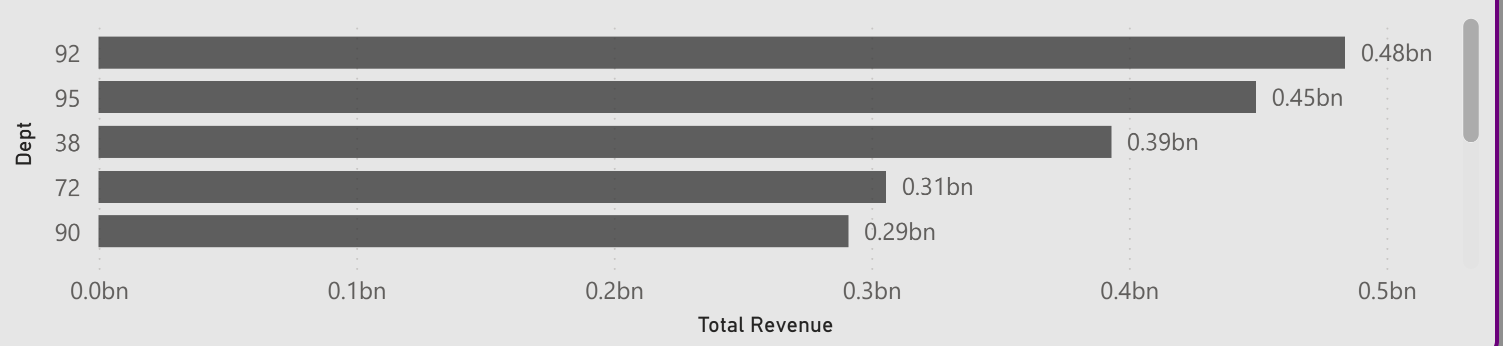




Store & Department Analysis

Top 10 Stores by Revenue

Top 10 Departments by Revenue



Revenue by Store Type

Year

- 2010
- 2011
- 2012

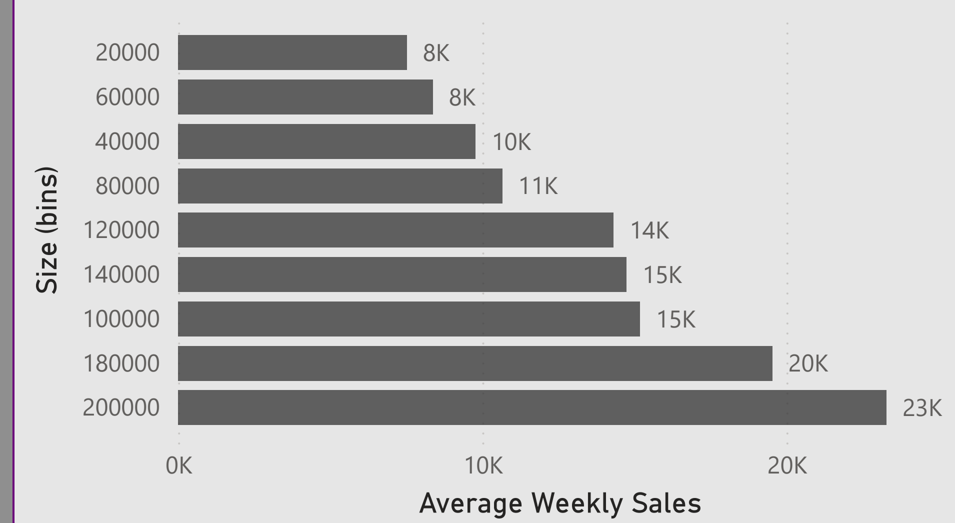
Stores

All

Department

All

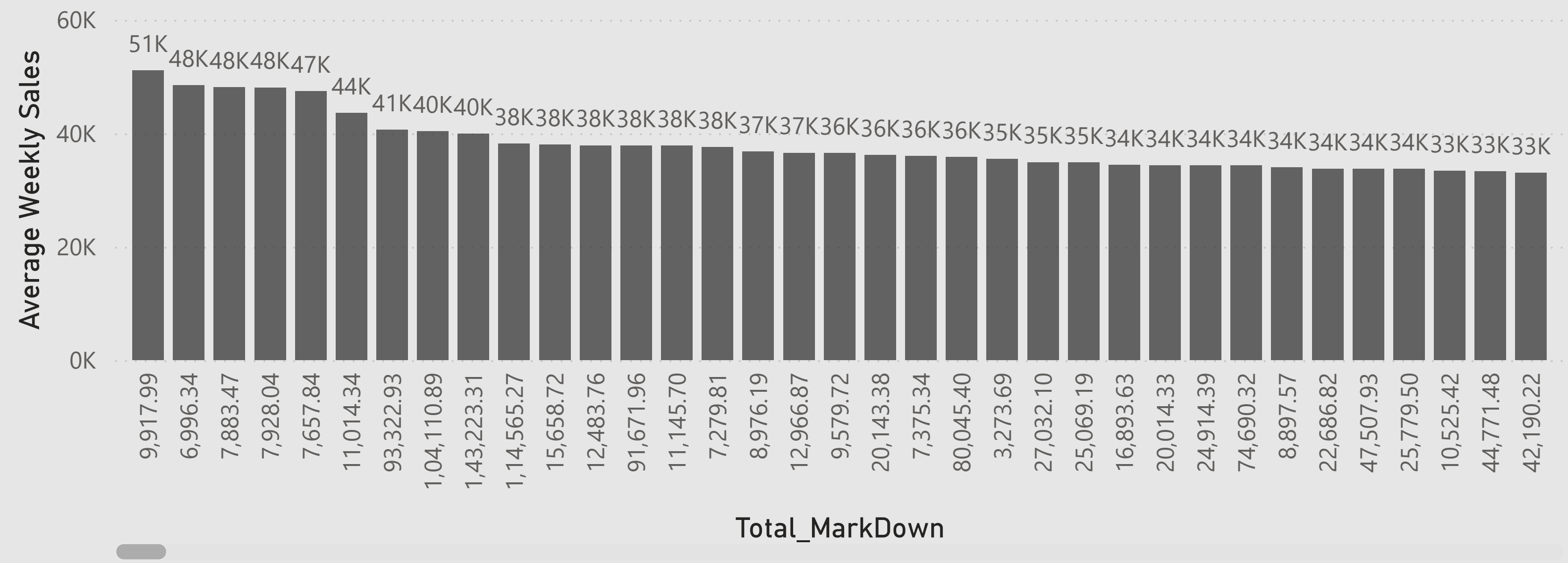
Average Weekly Sales by Store Size



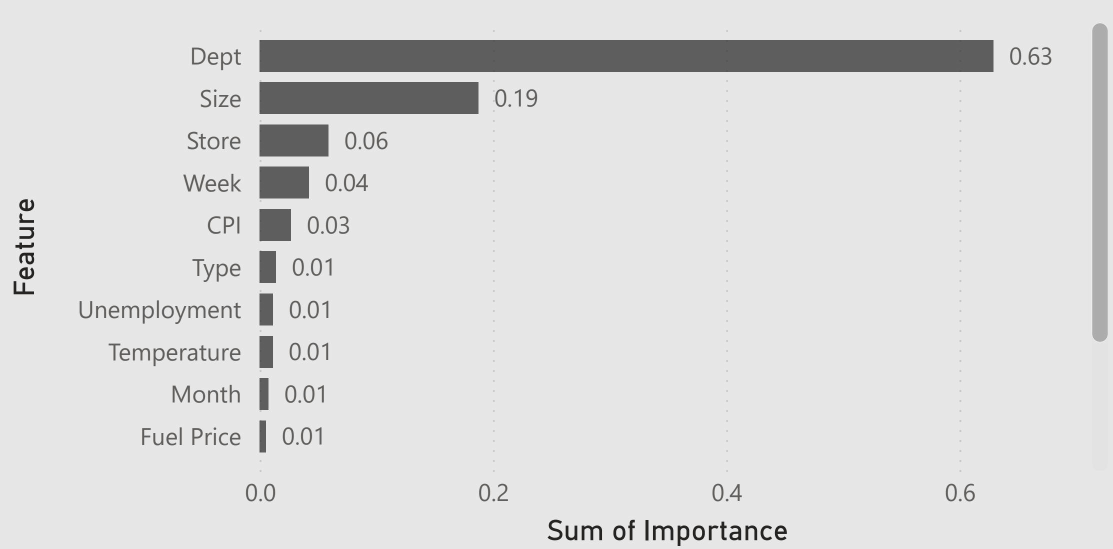
Sales Drivers & Business Insights

Holiday vs Non-Holiday Sales

Promotion Impact on Sales



Feature Importance (Random Forest)



Key Business Insights :

- Department is the primary driver of weekly sales (62.9% importance).
- Larger stores generally achieve higher sales than smaller stores.
- Sales performance varies significantly across stores, highlighting the need for store-specific planning.
- Weekly seasonal patterns have a noticeable impact on sales forecasting.
- Holidays and external economic factors have a relatively small influence on sales.
- The Random Forest model achieved an **R^2 score of 0.98**, making it suitable for accurate sales forecasting and inventory planning.

Sales Forecasting & Model Performance

Model Comparision

Model	MAE	R^2 Score	RMSE
Linear Regression	14,516.35	0.09	21,547.62
Random Forest	1,432.74	0.98	3,448.21

Random Forest
Best Model

R^2 Score
0.98

RMSE
3.45K

Strategic Recommendations :

- Prioritize inventory for high-performing departments.
- Allocate more resources to larger stores with consistently high sales.
- Use weekly forecasts to improve inventory and staffing decisions.
- Focus business planning on department-level demand rather than external economic factors.
- Deploy the Random Forest model for accurate weekly sales forecasting.

Project Summary :

- Performed end-to-end data cleaning and feature engineering.
- Built interactive Power BI dashboards for business analysis.
- Compared Linear Regression and Random Forest models.
- Random Forest achieved an R^2 score of 0.97, significantly outperforming Linear Regression.
- Developed business recommendations to support inventory planning and sales forecasting.